

PROJECT REPORT

Retail Business Performance & Profitability Analysis

Business Intelligence & Data Analytics

A comprehensive SQL · Python · Tableau analytics project examining category-level profitability, seasonal sales patterns, inventory turnover behavior, and strategic retail performance optimization.

- Tools Used: SQL (MySQL Workbench) | Python (Pandas, NumPy) | Tableau
- Dataset: 2,000 Retail Transaction Records | 5 Categories | 4 Regions
- Prepared: June 2026 | Internship Submission

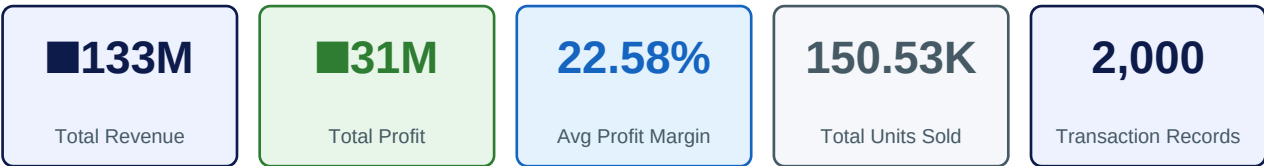
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01 — EXECUTIVE SUMMARY

This report presents the findings of a comprehensive Business Intelligence project conducted on a retail transactional dataset comprising **2,000 records** across five product categories, four geographic regions, and all four seasons. The analysis was executed using a structured data pipeline: raw data cleaning in **Python (Pandas)**, aggregation and business querying in **SQL (MySQL Workbench)**, and interactive dashboard development in **Tableau**.

The primary objective was to surface profit-driving and profit-draining segments, quantify seasonal demand patterns, and evaluate the relationship between inventory holding duration and overall profitability — enabling management to make data-backed strategic decisions.



■ Executive Highlights

- **Furniture** achieved the highest profit margin at 25.23%, driven by premium pricing and strong unit economics.
- **Clothing** recorded the lowest margin at 22.93%, signaling potential discount over-reliance.
- **Winter** generated the highest revenue (■44.6M); **Summer** produced the highest absolute profit.
- **Autumn** delivered the best seasonal margin at 24.77% — a key window for margin optimization.
- **Snacks, Tables, and Dairy** emerged as the top three sub-categories by profitability.
- Inventory Days and Profit show **virtually zero correlation** ($r = -0.007$), indicating turnover duration alone is not a reliable profit predictor.

The companion Tableau dashboard provides interactive exploration across Region, Season, and Category dimensions — empowering stakeholders to drill into specific market segments without requiring technical expertise.

02 — PROJECT OVERVIEW & OBJECTIVES

Retail businesses operate in highly competitive, margin-sensitive environments. Without systematic data analysis, decision-makers risk misallocating inventory, under-pricing high-value categories, and missing seasonal demand signals. This project was designed to eliminate that ambiguity through rigorous, multi-tool business intelligence analysis.

Project Objectives

- Identify profit-driving and profit-draining product categories and sub-categories.
- Quantify category and sub-category contribution to total revenue and profit.
- Evaluate seasonal revenue, profit, and margin performance across all four seasons.
- Examine the statistical relationship between inventory holding duration and profitability.
- Develop an interactive Tableau dashboard to support management decision-making.
- Generate actionable strategic recommendations for inventory, pricing, and seasonal planning.

Analytical Workflow

Stage	Tool	Activity	Output
1. Data Ingestion	Python / Pandas	Load raw CSV, inspect structure	Data profile report
2. Data Cleaning	Python / Pandas	Handle nulls, validate types, export	retail_data_clean.csv
3. Data Import	MySQL Workbench	Load cleaned CSV into SQL schema	retail_data_clean table
4. SQL Analysis	SQL (MySQL)	Aggregations, groupings, margins	Business insights
5. Visualization	Tableau Desktop	Dashboard with KPIs and charts	Interactive dashboard
6. Reporting	Python (ReportLab)	Compile findings into PDF report	This report

Tools & Technologies

SQL

MySQL Workbench
Aggregation & Querying

Python

Pandas & NumPy Data
Cleaning & Stats

Tableau

Dashboard Design Interactive
Visuals

CSV Data

2,000 Records Retail
Transactions

03 — DATASET DESCRIPTION

The dataset captures retail transactional records across multiple product categories, geographic regions, and seasonal periods. Each record represents a single transaction line, containing both operational (units, pricing, discount) and derived financial metrics (revenue, profit, margin).

Data Dictionary

Field	Data Type	Description	Business Relevance
order_id	INTEGER	Unique transaction ID	Record traceability
category	VARCHAR	Top-level product category (5 groups)	Category segmentation
sub_category	VARCHAR	Product sub-type (20 groups)	SKU-level analysis
region	VARCHAR	Geographic region (N/S/E/W)	Regional performance
season	VARCHAR	Season of sale (4 seasons)	Demand seasonality
sale_price	DECIMAL	Listed unit price before discount	Pricing strategy
cost_price	DECIMAL	Unit procurement cost	Cost management
units_sold	INTEGER	Quantity transacted	Volume analysis
inventory_days	INTEGER	Days held in inventory before sale	Turnover efficiency
discount_pct	DECIMAL	Discount rate applied (0.00–0.30)	Promotion analysis
revenue	DECIMAL	Effective revenue after discount	Top-line performance
cogs	DECIMAL	Total cost of goods sold	Cost of sales
profit	DECIMAL	Revenue minus COGS	Bottom-line health
profit_margin_pct	DECIMAL	Profit as % of Revenue	Profitability KPI

2,000

Raw Records

5

Product Categories

20

Sub-Categories

4

Regions

4

Seasons

14

Data Fields

04 — DATA CLEANING PROCESS (PYTHON)

Before any analytical work, raw data was subjected to a structured cleaning pipeline using **Python and Pandas**. This step is critical to ensure that downstream SQL analysis and Tableau visualizations are built on reliable, statistically sound data.

Cleaning Steps Performed

Step	Action	Finding / Outcome
1	Load dataset	2,000 records loaded successfully
2	Inspect data types	All types validated — no type mismatches
3	Identify missing values	187 null records found across 3 columns
4	Check duplicate records	0 duplicates — dataset is unique
5	Impute inventory_days nulls	Replaced with column median
6	Impute discount_pct nulls	Replaced with column median
7	Recalculate profit_margin_pct nulls	Derived from profit / revenue × 100
8	Add inventory_status column	Fast Moving (≤60d) / Slow Moving (>60d)
9	Add profit_status column	Profit / Loss classification
10	Export cleaned dataset	Saved as retail_data_clean.csv

Why Median Imputation? (Not Zero-Fill)

■ Methodology Note: Median vs Zero Imputation

- Replacing missing numerical values with **zero** would artificially suppress revenue, profit, and inventory metrics — creating statistical outliers that distort all aggregate calculations.
- The **median** is robust to outliers (unlike the mean) and reflects the central tendency of each column, ensuring imputed values are plausible within the real data distribution.
- For **inventory_days**, which frequently contains outliers from slow-moving stock, median imputation prevents extreme values from skewing turnover analysis.
- For **profit_margin_pct**, re-derivation from existing revenue and profit fields is preferred over any statistical imputation — preserving mathematical consistency.

Python Code — Data Cleaning Script

```
import pandas as pd

# Load raw dataset
df = pd.read_csv("retail_data_raw.csv")
```

```
print(df.shape) # (2000, 14)

# Audit: missing values
print(df.isnull().sum())

# inventory_days: ~106 nulls
# discount_pct: ~107 nulls
# profit_margin_pct: ~88 nulls

# Check duplicates
print("Duplicates:", df.duplicated().sum()) # 0

# Impute with median (not zero)
df['inventory_days'] = df['inventory_days'].fillna(df['inventory_days'].median())
df['discount_pct'] = df['discount_pct'].fillna(df['discount_pct'].median())

# Recalculate profit_margin_pct where null
mask = df['profit_margin_pct'].isnull()
df.loc[mask, 'profit_margin_pct'] = (df.loc[mask, 'profit'] / df.loc[mask, 'revenue']) * 100

# Add derived classification columns
df['inventory_status'] = df['inventory_days'].apply(lambda x: 'Slow Moving' if x > 60 else 'Fast Moving')
df['profit_status'] = df['profit'].apply(lambda x: 'Loss' if x < 0 else 'Profit')

# Export cleaned data
df.to_csv("retail_data_clean.csv", index=False)
print("Cleaning complete. Records:", len(df)) # 2000

# Correlation: Inventory Days vs Profit
corr = df['inventory_days'].corr(df['profit'])
print("Correlation:", corr) # -0.007079730279715629
```

05 — SQL ANALYSIS & BUSINESS QUERIES

The cleaned dataset was imported into **MySQL Workbench** and a dedicated **retail_analysis** database was created. SQL was used to perform all business aggregations — enabling fast, reproducible querying across multiple analytical dimensions.

Database Setup

```
CREATE DATABASE retail_analysis;

USE retail_analysis;

-- Verify record count after import
SELECT COUNT(*) FROM retail_data_clean; -- 2000

-- Total business performance
SELECT
SUM(revenue) AS total_revenue,
SUM(profit) AS total_profit
FROM retail_data_clean;
```

Query 1 — Category-Wise Profitability

```
SELECT
category,
ROUND(SUM(revenue), 2) AS revenue,
ROUND(SUM(profit), 2) AS profit,
ROUND((SUM(profit) / SUM(revenue)) * 100, 2) AS profit_margin
FROM retail_data_clean
GROUP BY category
ORDER BY profit DESC;
```

Query 2 — Seasonal Performance Analysis

```
SELECT
season,
ROUND(SUM(revenue), 2) AS total_revenue,
ROUND(SUM(profit), 2) AS total_profit,
ROUND((SUM(profit) / SUM(revenue)) * 100, 2) AS profit_margin
FROM retail_data_clean
GROUP BY season
ORDER BY total_profit DESC;
```

Query 3 — Inventory vs Profitability

```
SELECT
sub_category,
```



```
ROUND(AVG(inventory_days), 2) AS avg_inventory_days,  
ROUND(SUM(profit), 2) AS total_profit  
FROM retail_data_clean  
GROUP BY sub_category  
ORDER BY avg_inventory_days DESC;
```

06 — CATEGORY PROFITABILITY ANALYSIS

Category-level analysis reveals the contribution of each product group to overall business performance. Profit margins were calculated as a percentage of category revenue to ensure comparability across categories with different price points and volumes.

Category	Total Revenue (M)	Total Profit (M)	Profit Margin %	Rank
Furniture	32,602,101.72	8,225,332.40	25.23%	#1 ▲
Sports	33,582,619.38	8,206,251.46	24.44%	#2
Food & Beverage	37,281,014.59	8,642,454.37	23.18%	#3
Electronics	35,464,442.14	8,164,973.56	23.02%	#4
Clothing	31,850,290.73	7,303,377.40	22.93%	#5 ▼



■ Key Insights

- **Furniture** leads all categories with a 25.23% margin — primarily due to premium pricing and lower discount rates relative to cost.
- **Food & Beverage** generates the highest absolute revenue (M37.3M), validating its role as the volume anchor of the business.
- **Sports** ranks 2nd in margin (24.44%) with strong unit economics across gym equipment and supplements.
- **Clothing** shows the lowest margin at 22.93%, suggesting the category absorbs heavier promotional discounting that compresses profitability.
- Overall category margins are clustered within a **2.3 percentage point range** (22.93%–25.23%), indicating relatively balanced but improvable cost structures.

07 — SEASONAL PERFORMANCE ANALYSIS

Seasonal analysis identifies demand cycles, peak revenue periods, and margin fluctuations across the retail calendar. Understanding seasonal behavior enables proactive inventory planning, staffing, and promotional scheduling.

Season	Total Revenue (■)	Total Profit (■)	Profit Margin %	Revenue Rank
Winter	44,645,304.18	10,533,206.70	23.59%	#1 — Highest Revenue
Summer	43,662,632.02	10,594,730.28	24.26%	#2 — Highest Profit
Spring	41,935,401.33	9,373,752.92	22.35%	#3
Autumn	40,537,131.03	10,040,699.29	24.77%	#4 — Best Margin



Strategic Seasonal Observations

Winter is the revenue powerhouse, driven by holiday gifting, festive electronics demand, and peak household spending. Inventory should be maximised for Electronics and Furniture entering November.

Summer generates the highest absolute profit despite slightly lower revenue than Winter — suggesting better cost control and stronger product mix (Sports, Outdoor gear) with less promotional pressure.

Autumn delivers the best margin (24.77%), indicating disciplined pricing and reduced discounting post-Summer clearance. This season represents an opportunity to rebuild margin after Spring erosion.

Spring is the weakest season across all metrics. Revenue and margins dip simultaneously, pointing to demand slack compounded by clearance discounting. This is the optimal window for inventory liquidation campaigns.

■ Key Insights

- Winter and Summer together account for approximately **52% of annual revenue** — front-loading inventory for these periods is essential.
- Spring's low margin (22.35%) is partly structural: post-Winter clearance discounts compress category-level margins.
- Autumn's margin premium (24.77%) creates an opportunity to introduce new lines at full price before Winter promotional cycles begin.
- Seasonal variation in profit margin spans **2.42 percentage points** — a meaningful spread that seasonal planning can narrow.

08 — SUB-CATEGORY PROFITABILITY ANALYSIS

Sub-category analysis enables SKU-level profitability decisions. While categories provide strategic direction, sub-categories identify specific product lines to prioritize for inventory investment, marketing spend, or pricing review.

Top 10 Sub-Categories by Profitability

Rank	Sub-Category	Parent Category	Business Action
1	Snacks	Food & Beverage	Increase stock depth & shelf visibility
2	Tables	Furniture	Expand product range; prioritize in Autumn
3	Dairy	Food & Beverage	Ensure supply chain reliability; high frequency
4	Laptops	Electronics	Peak demand in Winter — pre-build inventory
5	Lamps	Furniture	Bundle with furniture sets; upsell opportunity
6	Supplements	Sports	Consistent demand — maintain year-round stock
7	Cameras	Electronics	Summer & Wedding season focus
8	Accessories	Electronics	High-margin add-on; promote with primary SKUs
9	Womens Wear	Clothing	Premium range focus to offset category margin gap
10	Apparel	Sports	Strong Summer demand; seasonal promotional tie-in

Sub-Category Strategic Priorities

Food & Beverage (Snacks, Dairy): These sub-categories combine strong absolute profitability with fast inventory turnover — a rare combination in retail analytics. Operational priority should be maintaining supply chain reliability and expanding shelf depth for these lines.

Electronics (Laptops, Cameras, Accessories): High-value items with concentrated seasonal demand. Accessories particularly benefit from bundling strategies with primary electronics purchases, adding margin with minimal incremental cost.

Furniture (Tables, Lamps): These sub-categories anchor the Furniture category's #1 margin ranking. Autumn home-refresh campaigns and office furniture promotions can extend demand beyond the traditional peak.

■ Sub-Category Action Priorities

- Snacks and Dairy alone likely represent a disproportionate share of Food & Beverage profit — isolating and protecting these lines is a high-priority management action.
- Electronics accessories are a margin amplifier: low COGS, high perceived value, easily bundled with primary electronics purchases.
- Womens Wear is the Clothing sub-category most likely to sustain premium pricing — reducing blanket discount exposure here would improve category margins.

09 — INVENTORY VS PROFITABILITY ANALYSIS

One of the core analytical questions of this project was whether inventory holding duration — measured in days — is a reliable predictor of product profitability. This analysis was conducted using Python's Pandas library, computing the Pearson correlation coefficient between **inventory_days** and **profit**.

Python Correlation Analysis

```
# Pearson correlation between inventory holding duration and profit
corr = df['inventory_days'].corr(df['profit'])
print("Correlation coefficient:", corr)

# Output:
# Correlation coefficient: -0.007079730279715629
```

-0.007

Pearson r (Inventory × Profit)

~0%

Variance Explained by Inv.
Days

No

Linear Relationship Detected

Yes

Product-Level Analysis
Required

Interpretation & Business Implications

A Pearson r of **-0.007** is statistically indistinguishable from zero. This confirms that **there is virtually no linear relationship** between how long a product sits in inventory and how profitable it ultimately is. The scatter plot in the Tableau dashboard (Inventory Days vs Profitability) visually corroborates this finding — data points are uniformly distributed across all inventory durations without any discernible trend.

This is a **critical business insight**: retail managers should not assume that slow-moving products are unprofitable. A product with 120 inventory days may generate higher margins than one that turns in 10 days — particularly in premium Furniture and Electronics, where longer holding periods are expected and justified by per-unit profit.

■ Analytical Interpretation

- **Slow-moving ≠ unprofitable.** Inventory holding duration is not a standalone profitability signal.
- **Category context matters:** Furniture naturally holds longer but delivers the highest margin (25.23%). Food & Beverage turns fast but at lower per-unit profit.
- **Recommendation:** Use **profit per inventory day** as a composite metric rather than inventory days alone — this surfaces true carrying cost efficiency.
- The scatter plot in the Tableau dashboard confirms near-random distribution with no visible slope — supporting the $r = -0.007$ finding.
- Sub-category and product-level profit analysis is a far more reliable management tool than inventory turnover duration alone.

10 — TABLEAU DASHBOARD

An interactive Tableau dashboard was developed to make the analytical findings accessible to non-technical stakeholders. The dashboard enables dynamic exploration of retail performance across Region, Season, and Category dimensions through point-and-click filtering.

Dashboard Screenshot



Figure 1 — Retail Business Performance & Profitability Dashboard (Tableau)

Dashboard KPIs



Dashboard Components

Visualization	Chart Type	Business Value
Profit by Season	Bar Chart	Identifies peak and trough demand periods for inventory planning
Profit by Category	Horizontal Bar	Ranks categories by profit contribution; guides budget allocation
Profit by Sub-Category	Horizontal Bar	Pinpoints high-value SKUs for promotional and stocking priority
Inventory Days vs Profitability	Scatter Plot	Tests correlation hypothesis; confirms near-zero relationship
KPI Summary Cards	BAN Tiles	Instant performance snapshot for executive stakeholders
KEY INSIGHTS Panel	Text Summary	Narrative context alongside visual data for complete understanding

Interactive Filters

- **Region:** East / North / South / West — isolate geographic performance.
- **Season:** All four seasons — drill into demand cycles.

- **Category:** All five product categories — cross-filter all views simultaneously.

11 — BUSINESS RECOMMENDATIONS

The following recommendations are derived directly from the SQL, Python, and Tableau analysis findings. Each recommendation is prioritized by expected business impact and mapped to the supporting data insight.

R1	Increase Inventory Allocation for Furniture & Sports These two categories deliver the highest profit margins (25.23% and 24.44%). Increasing inventory depth — particularly ahead of Winter and Summer peaks — will maximize margin yield from the business's strongest product lines. Avoid blanket promotions in these categories.
R2	Prioritize Snacks and Dairy — Maximize Shelf Depth Snacks and Dairy rank #1 and #3 by profitability and exhibit fast inventory turnover. These are the ideal 'cash engine' SKUs. Operational focus should be supply chain reliability and shelf visibility, not discount programs.
R3	Review Clothing Pricing & Discount Strategy Clothing records the lowest margin at 22.93%. Audit current discount depth across sub-categories — particularly mass-market lines. Shift promotional budget toward Womens Wear, which has stronger premium pricing potential.
R4	Pre-Build Inventory for Winter Demand Spikes Winter is the highest revenue season (■44.6M). Electronics and Furniture sub-categories drive this peak. Begin procurement 10–12 weeks in advance to prevent stockouts and capitalize on full-price selling opportunities.
R5	Expand Successful Furniture Product Lines Tables and Lamps are top-5 sub-categories by profitability. Expanding the Furniture range with complementary lines (office accessories, storage solutions) during Autumn home-refresh cycles can compound margin gains.
R6	Manage Inventory Using Profit-Per-Day, Not Duration Alone Given $r = -0.007$ between inventory days and profit, raw holding duration is not a reliable management signal. Implement a Profit-Per-Inventory-Day composite KPI to separate genuinely non-performing SKUs from premium items that naturally carry longer holding periods.
R7	Spring Clearance Strategy — Liquidate Before Summer Peak Spring is the weakest season (22.35% margin). Structure a targeted markdown calendar in February-March to clear slow-moving Clothing and Electronics before Summer demand peaks, freeing working capital for high-margin restocking.

12 — CONCLUSION

This project successfully delivered a **complete, end-to-end Business Intelligence pipeline** for retail performance and profitability analysis — from raw data ingestion through to strategic decision-support.

What Was Achieved

The **Python data cleaning process** resolved 187 missing values across three key columns using statistically sound median imputation, ensuring downstream analysis was built on clean, reliable data. The decision to use median — rather than zero-fill — preserved the distributional integrity of profitability, discount, and inventory metrics.

SQL analysis in MySQL Workbench produced granular aggregations across category, sub-category, season, and region dimensions — revealing that Furniture leads on margin (25.23%), Clothing lags (22.93%), Winter dominates revenue, and Autumn delivers the strongest profit margin of any season.

The **Python correlation analysis** produced a definitive finding: with $r = -0.007$, inventory holding duration has virtually no predictive power over profitability. This insight challenges conventional assumptions and recommends a shift to composite metrics like profit-per-inventory-day for more precise slow-mover identification.

The **Tableau dashboard** packages all findings into an interactive, filter-driven visualization environment — enabling non-technical stakeholders to explore performance by Region, Season, and Category without requiring SQL or Python skills.

■ Closing Summary

- Furniture and Sports are the margin leaders — protect and expand their inventory positions.
- Snacks and Dairy are the profit-volume combination leaders — operational focus, not discounting.
- Clothing requires a pricing and discount strategy review to close its margin gap.
- Winter and Summer together represent over half of annual revenue — inventory planning must be seasonal.
- Inventory duration is NOT a reliable profitability signal — product-level analysis is essential.
- Data-driven decision-making, as demonstrated in this project, can materially improve retail profitability, inventory efficiency, and seasonal revenue performance.

This project demonstrates how a structured BI approach — combining Python for data engineering, SQL for business querying, and Tableau for stakeholder communication — enables retail organisations to move from gut-feel decisions to evidence-based strategy. The analytical framework developed here is scalable to larger datasets and adaptable to real-time data pipelines as the business grows.